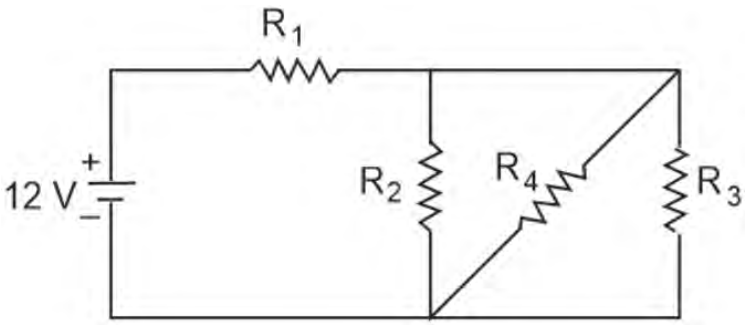
Q. 19 In the given figure $R_1 = 10\Omega, R_2 = 8\Omega, R_3 = 4\Omega$ and $R_4 = 8\Omega$. Battery is ideal with emf $12\ \text{V}$. Equivalent resistant of the circuit and current supplied by battery are respectively.



Option 1:
12Ω and $11.4\ \text{A}$

Option 2:
10.5Ω and $1.14\ \text{A}$

Option 3:
10.5Ω and $1\ \text{A}$

Option 4:
12Ω and $1\ \text{A}$

Correct Answer:
12Ω and $1\ \text{A}$

Solution:
Here $\text{R}_2, \text{R}_3, \text{R}_4$ are in parallel

$$\frac{1}{\text{R}_{234}} = \frac{1}{\text{R}_2} + \frac{1}{\text{R}_3} + \frac{1}{\text{R}_4}$$

$$\text{R}_{234} = 2\Omega$$

R_{234} is in series with R_1 so

$$\text{R}_{\text{eq}} = \text{R}_{234} + \text{R}_1 = 2 + 10 = 12\Omega$$

$\mathrm{i}=\frac{12}{12}=1\mathrm{Amp}$

Q. 20 An alternating voltage of amplitude $40\ \mathrm{V}$ and frequency $4\mathrm{kHz}$ is applied directly across the capacitor of $12\mu\mathrm{F}$. The maximum displacement current between the plates of the capacitor is nearly:

Option 1:
$13\ \mathrm{A}$

Option 2:
$8\ \mathrm{A}$

Option 3:
$10\ \mathrm{A}$

Option 4:
$12\ \mathrm{A}$

Correct Answer:
$12\ \mathrm{A}$

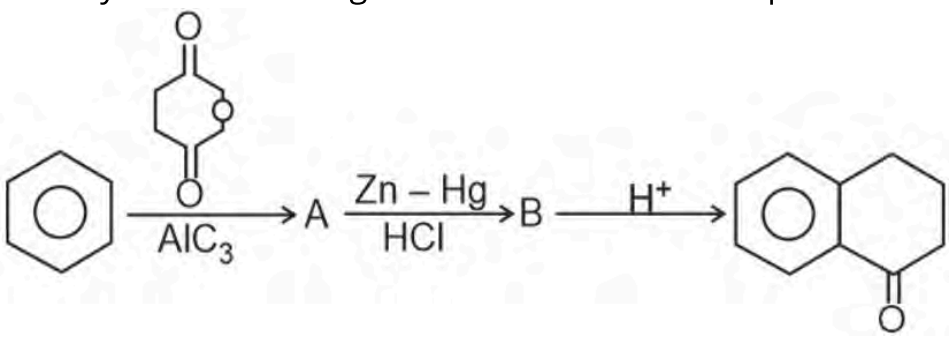
Solution:
Displacement current is same as conduction current in capacitor.

$X_C=\frac{1}{\omega \mathrm{C}}=\frac{1}{2\pi \mathrm{fC}}$

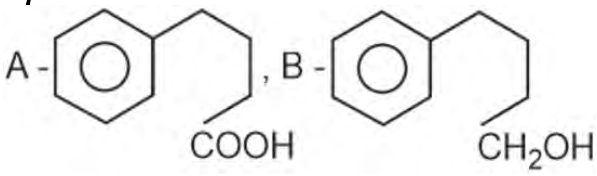
$=\frac{1}{2\pi\times4\times10^{3}\times12\times10^{-6}}=3.317\Omega$

$\mathrm{I}=\frac{\mathrm{V}}{\mathrm{X_C}}=\frac{40}{3.317}=12\ \mathrm{A}$

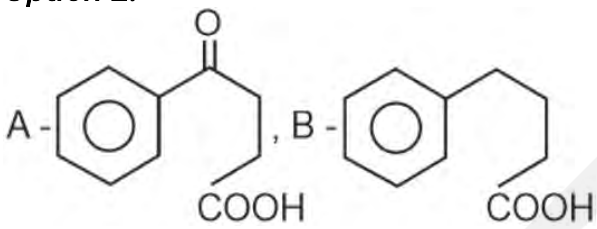
Q. 18 Identify A and B in the given chemical reaction sequence : -



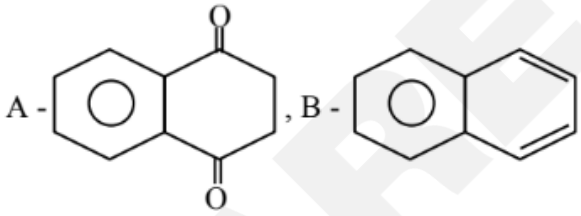
Option 1:



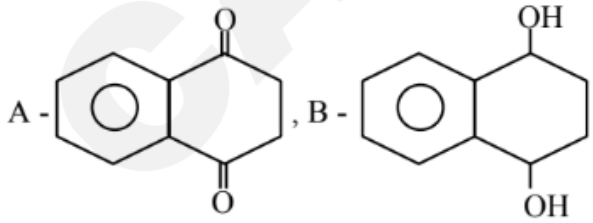
Option 2:



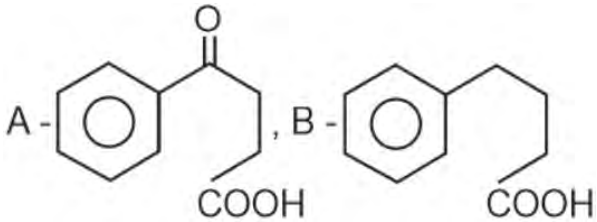
Option 3:



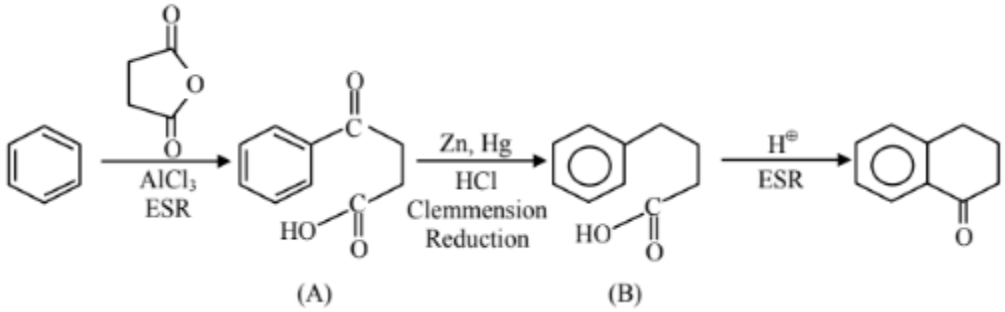
Option 4:



Correct Answer:



Solution:



Q. 19 The correct statements from the following are :

(A) The decreasing order of atomic radii of group 13 elements is $\mathrm{Tl} > \mathrm{In} > \mathrm{Ga} > \mathrm{Al} > \mathrm{B}$.

(B) Down the group 13 electronegativity decreases from top to bottom.

(C) Al dissolves in dil. HCl and liberate $\mathrm{H_2}$ but conc. $\mathrm{HNO_3}$ renders Al passive by forming a protective oxide layer on the surface.

(D) All elements of group 13 exhibits highly stable +1 oxidation state.

(E) Hybridisation of Al in $[\mathrm{Al(H_2O)_6}]^{3+}$ ion is $\mathrm{sp^3\ d^2}$.

Choose the **correct** answer from the options given below :

Option 1:
(C) and (E) only

Option 2:
(A), (C) and (E) only

Option 3:
(A), (B), (C) and (E) only

Option 4:
(A) and (C) only

Correct Answer:
(C) and (E) only

Solution:
A. size order $\mathrm{T\ell} > \mathrm{In} > \mathrm{Al} > \mathrm{Ga} > \mathrm{B}$
B. Electronegativity order $\mathrm{B} > \mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{T\ell}$
D. $\mathrm{B, Al}$ are more stable in +3 oxidation state

So, only C, E statements are correct.

Q. 23 For a differentiable function $f : IR \to IR$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in \text{IR}$, $f(0) = 1$ and $\lim_{x \to -\infty} f(x) = 7$. Then $9f(-\log_e 3)$ is equal to

Correct Answer:
61

Solution:

$$\frac{dy}{dx} - 3y = \alpha$$

$$\text{If } = e^{\int -3dx} = e^{-3x}$$

$$\therefore y - e^{-3x} = \int e^{-3x} \cdot \alpha dx$$

$$ye^{-3x} = \frac{\alpha e^{-3x}}{-3} + c$$

$$\left(*e^{3x}\right)$$

$$y = \frac{\alpha}{-3} + C \cdot e^{3x}$$

$$\text{on substituting } x = 0, y = 1$$

$$x \to -\infty, y = 7$$

we get $y = 7 - 6e^{3x}$

$$9f(-\log_e 3) = 61$$

Q. 24 The number of integers, between 100 and 1000 having the sum of their digits equals to 14 , is

Correct Answer:
70

Solution:

$$\text{N} = \text{abc}$$

(i) All distinct digits

$$a + b + c = 14$$

$$a \geq 1$$

$$b, c \in \{0 \text{ to } 9\}$$

by hit & trial : 8 cases

$(6,5,3)$ $(8,6,0)$ $(9,4,1)$

$(7,6,1)$ $(8,5,1)$ $(9,3,2)$

$(7,5,2)$ $(8,4,2)$

$(7,4,3)$ $(9,5,0)$

(ii) 2 same, 1 diff $a = b; c$

$2a + c = 14$

by values :

$(3,8)$ $(4,6)$ $(5,4)$ $(6,2)$ $(7,0)$ $\Big\} \frac{3!}{2!} \times 5 - 1$

$= 14$ cases

(iii) all same :

$3a = 14$

$a = \frac{14}{3} \times$ rejected

0 cases

Hence, Total cases :

$8 \times 3! + 2 \times (4) + 14$

$= 48 + 22$

$= 70$

Q. 25 Let $\mathrm{A} = \{(\mathrm{x}, \mathrm{y}) : 2\mathrm{x} + 3\mathrm{y} = 23, \mathrm{x}, \mathrm{y} \in \mathrm{N}\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to

Correct Answer:

24

Solution: